1 Discrete Mathematics: Combinatory (50)

Syllabus: Combinatorics: Counting, Recurrence relations, Generating functions.

Mark Distribution in Previous GATE

Year	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum	Average	Maximum
1 Mark Count	1	0	0	0	1	1	1	0	0	0.5	1
2 Marks Count	0	0	0	0	1	2	0	1	0	0.5	2
Total Marks	1	0	0	0	3	5	1	2	**0**	**1.5**	**5**

1.1 Balls In Bins (4)

1.1.1 Balls In Bins: GATE CSE 2002 | Question: 13

a. In how many ways can a given positive integer $n \geq 2$ be expressed as the sum of 2 positive integers (which are not necessarily distinct). For example, for $n = 3$, the number of ways is 2, i.e., $1+2, 2+1$. Give only the answer without any explanation.
b. In how many ways can a given positive integer $n \geq 3$ be expressed as the sum of 3 positive integers (which are not necessarily distinct). For example, for $n = 4$, the number of ways is 3, i.e., $1+2+1, 2+1+1$ and $1+1+2$. Give only the answer without explanation.
c. In how many ways can a given positive integer $n \geq k$ be expressed as the sum of k positive integers (which are not necessarily distinct). Give only the answer without explanation.

gatecse-2002 combinatory normal descriptive balls-in-bins

Answer key☟

1.1.2 Balls In Bins: GATE CSE 2003 | Question: 34

m identical balls are to be placed in n distinct bags. You are given that $m \geq kn$, where k is a natural number ≥ 1. In how many ways can the balls be placed in the bags if each bag must contain at least k balls?

A. $\begin{pmatrix} m-k \\ n-1 \end{pmatrix}$
B. $\begin{pmatrix} m-kn+n-1 \\ n-1 \end{pmatrix}$
C. $\begin{pmatrix} m-1 \\ n-k \end{pmatrix}$
D. $\begin{pmatrix} m-kn+n+k-2 \\ n-k \end{pmatrix}$

gatecse-2003 combinatory balls-in-bins normal

Answer key☟

1.1.3 Balls In Bins: GATE CSE 2022 | Question: 22

The number of arrangements of six identical balls in three identical bins is ______________ .

gatecse-2022 numerical-answers combinatory balls-in-bins one-mark

Answer key☟

1.1.4 Balls In Bins: GATE IT 2004 | Question: 35

In how many ways can we distribute 5 distinct balls, $B_1, B_2, \ldots, B_5$ in 5 distinct cells, $C_1, C_2, \ldots, C_5$ such that Ball B_i is not in cell C_i, $\forall i = 1, 2, \ldots 5$ and each cell contains exactly one ball?

A. 44
B. 96
C. 120
D. 3125

gateit-2004 combinatory normal balls-in-bins

Answer key☟

1.2 Combinatory (22)

1.2.1 Combinatory: GATE CSE 1989 | Question: 4-i

How many substrings (of all lengths inclusive) can be formed from a character string of length n? Assume all characters to be distinct, prove your answer.

gate1989 descriptive combinatory normal proof

Answer key

1.2.2 Combinatory: GATE CSE 1990 | Question: 3-iii

The number of rooted binary trees with n nodes is,

A. Equal to the number of ways of multiplying $(n+1)$ matrices.
B. Equal to the number of ways of arranging n out of $2n$ distinct elements.
C. Equal to $\frac{1}{(n+1)}\binom{2n}{n}$.
D. Equal to $n!$.

gate1990 normal combinatory catalan-number multiple-selects

Answer key

1.2.3 Combinatory: GATE CSE 1990 | Question: 3-ix

The number of ways in which $5\ A's, 5\ B's$ and $5\ C's$ can be arranged in a row is:

A. $15!/(5!)^3$ B. $15!$ C. $\binom{15}{5}$ D. $15!(5!3!)$.

gate1990 normal combinatory

Answer key

1.2.4 Combinatory: GATE CSE 1991 | Question: 02-iv

Match the pairs in the following questions by writing the corresponding letters only.

$A.$	The number of distinct binary tree with n nodes.	$P.$	$\frac{n!}{2}$
$B.$	The number of binary strings of the length of 2n with an equal number of 0's and 1's	$Q.$	$\binom{3n}{n}$
$C.$	The number of even permutation of n objects.	$R.$	$\binom{2n}{n}$
$D.$	The number of binary strings of length 6n which are palindromes with 2n 0's.	$S.$	$\frac{1}{1+n}\binom{2n}{n}$

gate1991 combinatory normal match-the-following

Answer key

1.2.5 Combinatory: GATE CSE 1991 | Question: 16,a

Find the number of binary strings w of length $2n$ with an equal number of $1's$ and $0's$ and the property that every prefix of w has at least as many $0's$ as $1's$.

gate1991 combinatory normal descriptive catalan-number

Answer key

1.2.6 Combinatory: GATE CSE 1998 | Question: 1.23

How many sub strings of different lengths (non-zero) can be formed from a character string of length n?

A. n B. n^2 C. 2^n D. $\frac{n(n+1)}{2}$

gate1998 combinatory normal

Answer key

1.2.7 Combinatory: GATE CSE 1999 | Question: 1.3

The number of binary strings of n zeros and k ones in which no two ones are adjacent is

A. ${}^{n-1}C_k$ B. ${}^{n}C_k$ C. ${}^{n}C_{k+1}$ D. None of the above

gate1999 combinatory normal

Answer key☟

1.2.8 Combinatory: GATE CSE 1999 | Question: 2.2

Two girls have picked 10 roses, 15 sunflowers and 15 daffodils. What is the number of ways they can divide the flowers among themselves?

A. 1638 B. 2100 C. 2640 D. None of the above

gate1999 combinatory normal

Answer key☟

1.2.9 Combinatory: GATE CSE 2000 | Question: 5

A multiset is an unordered collection of elements where elements may repeat any number of times. The size of a multiset is the number of elements in it, counting repetitions.

a. What is the number of multisets of size 4 that can be constructed from n distinct elements so that at least one element occurs exactly twice?
b. How many multisets can be constructed from n distinct elements?

gatecse-2000 combinatory normal descriptive

Answer key☟

1.2.10 Combinatory: GATE CSE 2001 | Question: 2.1

How many 4-digit even numbers have all 4 digits distinct?

A. 2240 B. 2296 C. 2620 D. 4536

gatecse-2001 combinatory normal

Answer key☟

1.2.11 Combinatory: GATE CSE 2003 | Question: 4

Let A be a sequence of 8 distinct integers sorted in ascending order. How many distinct pairs of sequences, B and C are there such that

i. each is sorted in ascending order,
ii. B has 5 and C has 3 elements, and
iii. the result of merging B and C gives A

A. 2 B. 30 C. 56 D. 256

gatecse-2003 combinatory normal

Answer key☟

1.2.12 Combinatory: GATE CSE 2003 | Question: 5

n couples are invited to a party with the condition that every husband should be accompanied by his wife. However, a wife need not be accompanied by her husband. The number of different gatherings possible at the party is

A. ${}^{2n}\mathrm{C}_n \times 2^n$ B. 3^n C. $\frac{(2n)!}{2^n}$ D. ${}^{2n}\mathrm{C}_n$

gatecse-2003 combinatory normal

Answer key☟

1.2.13 Combinatory: GATE CSE 2004 | Question: 75

Mala has the colouring book in which each English letter is drawn two times. She wants to paint each of these 52 prints with one of k colours, such that the colour pairs used to colour any two letters are different. Both prints of a letter can also be coloured with the same colour. What is the minimum value of k that satisfies this requirement?

A. 9 B. 8 C. 7 D. 6

gatecse-2004 combinatory

Answer key

1.2.14 Combinatory: GATE CSE 2007 | Question: 84

Suppose that a robot is placed on the Cartesian plane. At each step it is allowed to move either one unit up or one unit right, i.e., if it is at (i, j) then it can move to either $(i+1, j)$ or $(i, j+1)$.

How many distinct paths are there for the robot to reach the point $(10, 10)$ starting from the initial position $(0, 0)$?

A. ${}^{20}\mathrm{C}_{10}$ B. 2^{20} C. 2^{10} D. None of the above

gatecse-2007 combinatory

Answer key

1.2.15 Combinatory: GATE CSE 2007 | Question: 85

Suppose that a robot is placed on the Cartesian plane. At each step it is allowed to move either one unit up or one unit right, i.e., if it is at (i, j) then it can move to either $(i+1, j)$ or $(i, j+1)$.

Suppose that the robot is not allowed to traverse the line segment from $(4, 4)$ to $(5, 4)$. With this constraint, how many distinct paths are there for the robot to reach $(10, 10)$ starting from $(0, 0)$?

A. 2^9
B. 2^{19}
C. ${}^{8}\mathrm{C}_4 \times^{11} \mathrm{C}_5$
D. ${}^{20}\mathrm{C}_{10} -^{8} \mathrm{C}_4 \times^{11} \mathrm{C}_5$

gatecse-2007 combinatory normal discrete-mathematics

Answer key

1.2.16 Combinatory: GATE CSE 2014 Set 1 | Question: 49

A pennant is a sequence of numbers, each number being 1 or 2. An $n-$pennant is a sequence of numbers with sum equal to n. For example, $(1, 1, 2)$ is a $4-$pennant. The set of all possible $1-$pennants is (1), the set of all possible $2-$pennants is $(2), (1, 1)$ and the set of all $3-$pennants is $(2, 1), (1, 1, 1), (1, 2)$. Note that the pennant $(1, 2)$ is not the same as the pennant $(2, 1)$. The number of $10-$pennants is________

gatecse-2014-set1 combinatory numerical-answers normal

Answer key

1.2.17 Combinatory: GATE CSE 2015 Set 3 | Question: 5

The number of 4 digit numbers having their digits in non-decreasing order (from left to right) constructed by using the digits belonging to the set $\{1, 2, 3\}$ is ________.

gatecse-2015-set3 combinatory normal numerical-answers

Answer key

1.2.18 Combinatory: GATE CSE 2019 | Question: 5

Let $U = \{1, 2, \ldots, n\}$ Let $A = \{(x, X) \mid x \in X, X \subseteq U\}$. Consider the following two statements on $|A|$.

I. $|A| = n2^{n-1}$
II. $|A| = \Sigma_{k=1}^{n} k \binom{n}{k}$

Which of the above statements is/are TRUE?

A. Only I B. Only II C. Both I and II D. Neither I nor II

gatecse-2019 engineering-mathematics discrete-mathematics combinatory one-mark

Answer key

1.2.19 Combinatory: GATE CSE 2020 | Question: 42

The number of permutations of the characters in LILAC so that no character appears in its original position, if the two L's are indistinguishable, is ______.

gatecse-2020 numerical-answers combinatory two-marks

Answer key

1.2.20 Combinatory: GATE CSE 2025 | Set 1 | Question: 20

Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, "aa", "bb" or "cc". The number of such strings of length 5 that are possible is __________. (Answer in integer).

gatecse2025-set1 combinatory numerical-answers one-mark

Answer key

1.2.21 Combinatory: GATE IT 2005 | Question: 46

A line L in a circuit is said to have a $stuck-at-0$ fault if the line permanently has a logic value 0. Similarly a line L in a circuit is said to have a $stuck-at-1$ fault if the line permanently has a logic value 1. A circuit is said to have a multiple $stuck-at$ fault if one or more lines have stuck at faults. The total number of distinct multiple $stuck-at$ faults possible in a circuit with N lines is

A. 3^N B. 3^N-1 C. 2^N-1 D. 2

gateit-2005 combinatory normal

Answer key

1.2.22 Combinatory: GATE IT 2008 | Question: 25

In how many ways can b blue balls and r red balls be distributed in n distinct boxes?

A. $\dfrac{(n+b-1)!\,(n+r-1)!}{(n-1)!\,b!\,(n-1)!\,r!}$

B. $\dfrac{(n+(b+r)-1)!}{(n-1)!\,(n-1)!\,(b+r)!}$

C. $\dfrac{n!}{b!\,r!}$

D. $\dfrac{(n+(b+r)-1)!}{n!\,(b+r-1)}$

gateit-2008 combinatory normal

Answer key

1.3 Counting (4)

1.3.1 Counting: GATE CSE 1994 | Question: 1.15

The number of substrings (of all lengths inclusive) that can be formed from a character string of length n is

A. n B. n^2 C. $\frac{n(n-1)}{2}$ D. $\frac{n(n+1)}{2}$

gate1994 combinatory counting normal

Answer key

1.3.2 Counting: GATE CSE 2021 Set 1 | Question: 19

There are 6 jobs with distinct difficulty levels, and 3 computers with distinct processing speeds. Each job is assigned to a computer such that:

- The fastest computer gets the toughest job and the slowest computer gets the easiest job.
- Every computer gets at least one job.

The number of ways in which this can be done is ___________.

gatecse-2021-set1 combinatory counting numerical-answers one-mark

Answer key☟

1.3.3 Counting: GATE CSE 2021 Set 2 | Question: 50

Let S be a set of consisting of 10 elements. The number of tuples of the form (A, B) such that A and B are subsets of S, and $A \subseteq B$ is ___________

gatecse-2021-set2 combinatory counting numerical-answers two-marks

Answer key☟

1.3.4 Counting: GATE CSE 2023 | Question: 38

Let $U = \{1, 2, \ldots, n\}$, where n is a large positive integer greater than 1000. Let k be a positive integer less than n. Let A, B be subsets of U with $|A| = |B| = k$ and $A \cap B = \emptyset$. We say that a permutation of U separates A from B if one of the following is true.

- All members of A appear in the permutation before any of the members of B.
- All members of B appear in the permutation before any of the members of A.

How many permutations of U separate A from B?

A. $n!$

B. $\binom{n}{2k}(n-2k)!$

C. $\binom{n}{2k}(n-2k)!(k!)^2$

D. $2\binom{n}{2k}(n-2k)!(k!)^2$

gatecse-2023 combinatory counting two-marks

Answer key☟

1.4 Generating Functions (6)

1.4.1 Generating Functions: GATE CSE 1987 | Question: 10b

What is the generating function $G(z)$ for the sequence of Fibonacci numbers?

gate1987 combinatory generating-functions descriptive

Answer key☟

1.4.2 Generating Functions: GATE CSE 2005 | Question: 50

Let $G(x) = \frac{1}{(1-x)^2} = \sum\limits_{i=0}^{\infty} g(i)x^i$, where $|x| < 1$. What is $g(i)$?

A. i

B. $i + 1$

C. $2i$

D. 2^i

gatecse-2005 normal generating-functions

Answer key☟

1.4.3 Generating Functions: GATE CSE 2016 Set 1 | Question: 26

The coefficient of x^{12} in $\left(x^3 + x^4 + x^5 + x^6 + \ldots\right)^3$ is ___________.

gatecse-2016-set1 combinatory generating-functions normal numerical-answers

Answer key☟

1.4.4 Generating Functions: GATE CSE 2017 Set 2 | Question: 47

If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to ___________ .

gatecse-2017-set2 combinatory generating-functions numerical-answers normal

Answer key☟

1.4.5 Generating Functions: GATE CSE 2018 | Question: 1

Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0, 1, 2, \ldots$?

A. $\frac{3}{(1-x)^2}$ B. $\frac{3x}{(1-x)^2}$ C. $\frac{2-x}{(1-x)^2}$ D. $\frac{3-x}{(1-x)^2}$

gatecse-2018 generating-functions normal combinatory one-mark

Answer key☟

1.4.6 Generating Functions: GATE CSE 2022 | Question: 26

Which one of the following is the closed form for the generating function of the sequence $\{a_n\}_{n\geq 0}$ defined below?

$$a_n = \begin{cases} n+1, & \text{n is odd} \\ 1, & \text{otherwise} \end{cases}$$

A. $\frac{x(1+x^2)}{(1-x^2)^2} + \frac{1}{1-x}$ B. $\frac{x(3-x^2)}{(1-x^2)^2} + \frac{1}{1-x}$
C. $\frac{2x}{(1-x^2)^2} + \frac{1}{1-x}$ D. $\frac{x}{(1-x^2)^2} + \frac{1}{1-x}$

gatecse-2022 combinatory generating-functions two-marks

Answer key☟

1.5 Modular Arithmetic (2)

1.5.1 Modular Arithmetic: GATE CSE 2016 Set 2 | Question: 29

The value of the expression $13^{99} \pmod{17}$ in the range 0 to 16, is ________.

gatecse-2016-set2 modular-arithmetic normal numerical-answers

Answer key☟

1.5.2 Modular Arithmetic: GATE CSE 2019 | Question: 21

The value of $3^{51} \bmod 5$ is _____

gatecse-2019 numerical-answers combinatory modular-arithmetic one-mark

Answer key☟

1.6 Pigeonhole Principle (2)

1.6.1 Pigeonhole Principle: GATE CSE 2000 | Question: 1.1

The minimum number of cards to be dealt from an arbitrarily shuffled deck of 52 cards to guarantee that three cards are from same suit is

A. 3 B. 8 C. 9 D. 12

gatecse-2000 easy pigeonhole-principle combinatory

Answer key☟

1.6.2 Pigeonhole Principle: GATE CSE 2005 | Question: 44

What is the minimum number of ordered pairs of non-negative numbers that should be chosen to ensure that there are two pairs (a, b) and (c, d) in the chosen set such that, $a \equiv c \mod 3$ and $b \equiv d \mod 5$

A. 4 B. 6 C. 16 D. 24

gatecse-2005 set-theory&algebra normal pigeonhole-principle

Answer key☟

1.7 Recurrence Relation (7)

1.7.1 Recurrence Relation: GATE CSE 1996 | Question: 9

The Fibonacci sequence $\{f_1, f_2, f_3 \dots f_n\}$ is defined by the following recurrence:

$$f_{n+2} = f_{n+1} + f_n, n \geq 1; f_2 = 1 : f_1 = 1$$

Prove by induction that every third element of the sequence is even.

gate1996 recurrence-relation proof descriptive

Answer key

1.7.2 Recurrence Relation: GATE CSE 2016 Set 1 | Question: 2

Let a_n be the number of n-bit strings that do **NOT** contain two consecutive $1's$. Which one of the following is the recurrence relation for a_n?

A. $a_n = a_{n-1} + 2a_{n-2}$
B. $a_n = a_{n-1} + a_{n-2}$
C. $a_n = 2a_{n-1} + a_{n-2}$
D. $a_n = 2a_{n-1} + 2a_{n-2}$

gatecse-2016-set1 combinatory recurrence-relation easy

Answer key

1.7.3 Recurrence Relation: GATE CSE 2016 Set 1 | Question: 27

Consider the recurrence relation $a_1 = 8, a_n = 6n^2 + 2n + a_{n-1}$. Let $a_{99} = K \times 10^4$. The value of K is __________.

gatecse-2016-set1 combinatory recurrence-relation normal numerical-answers

Answer key

1.7.4 Recurrence Relation: GATE CSE 2022 | Question: 41

Consider the following recurrence:

$$\begin{aligned} f(1) &= 1; \\ f(2n) &= 2f(n)-1, \quad \text{for } n \geq 1; \\ f(2n+1) &= 2f(n)+1, \quad \text{for } n \geq 1. \end{aligned}$$

Then, which of the following statements is/are TRUE?

A. $f(2^n - 1) = 2^n - 1$
B. $f(2^n) = 1$
C. $f(5 \cdot 2^n) = 2^{n+1} + 1$
D. $f(2^n + 1) = 2^n + 1$

gatecse-2022 combinatory recurrence-relation multiple-selects two-marks

Answer key

1.7.5 Recurrence Relation: GATE CSE 2023 | Question: 5

The Lucas sequence L_n is defined by the recurrence relation:

$$L_n = L_{n-1} + L_{n-2}, \quad \text{for} \quad n \geq 3,$$

with $L_1 = 1$ and $L_2 = 3$.
Which one of the options given is TRUE?

A. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$
B. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{3}\right)^n$
C. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{3}\right)^n$
D. $L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n$

gatecse-2023 combinatory recurrence-relation one-mark

Answer key

1.7.6 Recurrence Relation: GATE IT 2004 | Question: 34

Let $H_1, H_2, H_3, ...$ be harmonic numbers. Then, for $n \in Z^+$, $\sum_{j=1}^{n} H_j$ can be expressed as

A. $nH_{n+1} - (n+1)$
B. $(n+1)H_n - n$
C. $nH_n - n$
D. $(n+1)H_{n+1} - (n+1)$

gateit-2004 recurrence-relation combinatory normal

Answer key

1.7.7 Recurrence Relation: GATE IT 2007 | Question: 76

Consider the sequence $\langle x_n \rangle$, $n \geq 0$ defined by the recurrence relation $x_{n+1} = c.x_n^2 - 2$, where $c > 0$.

Suppose there exists a ***non-empty, open*** interval (a, b) such that for all x_0 satisfying $a < x_0 < b$, the sequence converges to a limit. The sequence converges to the value?

A. $\frac{1+\sqrt{1+8c}}{2c}$ B. $\frac{1-\sqrt{1+8c}}{2c}$ C. 2 D. $\frac{2}{2c-1}$

gateit-2007 combinatory normal recurrence-relation

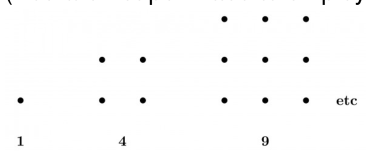
Answer key☟

1.8 Summation (3)

1.8.1 Summation: GATE CSE 1994 | Question: 15

Use the patterns given to prove that

A. $\sum\limits_{i=0}^{n-1}(2i+1)=n^2$
(You are not permitted to employ induction)



B. Use the result obtained in (A) to prove that $\sum\limits_{i=1}^{n} i=\frac{n(n+1)}{2}$

gate1994 combinatory proof summation descriptive

Answer key☟

1.8.2 Summation: GATE CSE 2008 | Question: 24

Let $P=\sum\limits_{\substack{1\leq i\leq 2k\\ i \text{ odd}}} i$ and $Q=\sum\limits_{\substack{1\leq i\leq 2k\\ i \text{ even}}} i$, where k is a positive integer. Then

A. $P=Q-k$ B. $P=Q+k$ C. $P=Q$ D. $P=Q+2k$

gatecse-2008 combinatory easy summation

Answer key☟

1.8.3 Summation: GATE CSE 2015 Set 1 | Question: 26

$\sum\limits_{x=1}^{99}\frac{1}{x(x+1)}$ = ________.

gatecse-2015-set1 combinatory normal numerical-answers summation

Answer key☟

Answer Keys

1.1.1	N/A	1.1.2	B	1.1.3	7	1.1.4	A	1.2.1	N/A
1.2.2	A;C	1.2.3	A	1.2.4	N/A	1.2.5	N/A	1.2.6	D
1.2.7	D	1.2.8	D	1.2.9	N/A	1.2.10	B	1.2.11	C
1.2.12	B	1.2.13	C	1.2.14	A	1.2.15	D	1.2.16	89
1.2.17	15	1.2.18	C	1.2.19	12	1.2.20	195:195	1.2.21	B
1.2.22	A	1.3.1	D	1.3.2	65 : 65	1.3.3	59049 : 59049	1.3.4	D
1.4.1	N/A	1.4.2	B	1.4.3	10	1.4.4	15	1.4.5	D
1.4.6	A	1.5.1	4	1.5.2	2	1.6.1	C	1.6.2	C
1.7.1	N/A	1.7.2	B	1.7.3	197.9 : 198.1	1.7.4	A;B;C	1.7.5	A
1.7.6	B	1.7.7	B	1.8.1	N/A	1.8.2	A	1.8.3	0.99